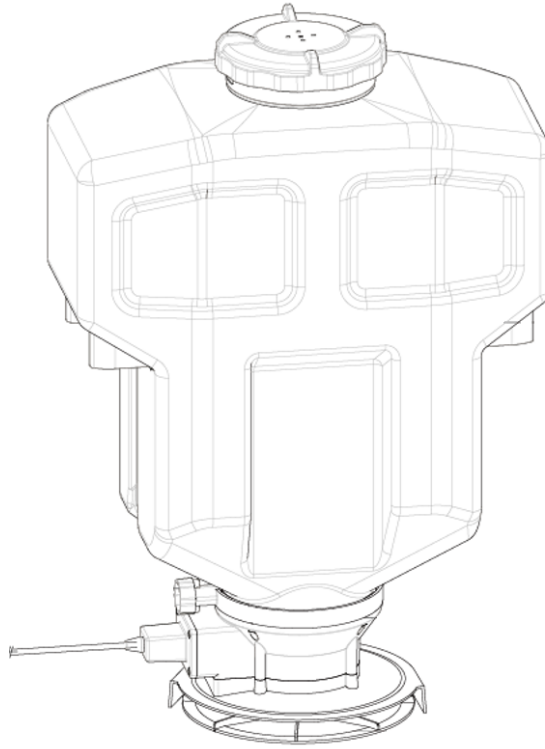


T Series Spreading System

Repair Guide For Dealers



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History				
Version	Content	Date	Written By	Approved By
V1.0	First Edition	2020.03.18	GKAS	
V1.1	Newly Added Assembly Notices before Aircraft Calibration	2020.07.01	GKAS	

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Purpose

- The purpose of this document is to show DJI authorized dealers how to repair the product, and it is only for dealers who have signed the confidentiality agreement and have a DJI product repair qualification. This document cannot be distributed to other third parties or be disclosed to the public. If you find any mistakes or have any suggestions, please contact the local DJI service manager.
- To find the related repair information and policies, please visit: <https://dealersystem.dji.com/>

Repair Notices

- Repair and calibration should be carried out by qualified technical staff.
- Please ensure that all operations are in an anti-static workshop.
- Please ensure that no soldering tin, screws, or debris are inside the product and its components.
- Please ensure that all modules, screws, and conductive foams are mounted in place and that glue is applied properly.
- Please ensure that all the cables are connected correctly.

Anti-static Protection

- Disassembly or assembly of the external shell must be done in an anti-static workshop.
- The electronic components are electrical static sensitive units. Electrostatic Discharge (ESD) is the main cause of damage to electronic components.
- Ensure that anti-static measures are taken when transporting the electrical static sensitive devices.
- Do not operate in a strong static or magnetic field.

Maintenance

- Keep the aircraft and other modules away from children.
- Do not let the camera come into contact with liquid or water.
- Do not power on the aircraft immediately after it has fallen into water.
- Do not store the camera in a humid area.

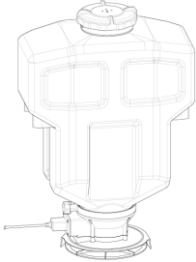
1 Product Introduction

The T Series Spreading System is compatible with Agras T series aircraft and offers efficient, reliable, and stable spreading operations. The material delivery system is precisely controlled by the built-in stirring device and hopper gate, which can prevent material blockages and improve operating accuracy and reliability.

Use the app compatible with your aircraft to set parameters such as the hopper outlet size and spinner disk rotating speed. These parameters can be adjusted to meet different requirements. The app provides warning prompts for an empty tank as well as for abnormalities in the rotating speed, temperature, and hopper outlet size. These prompts help ensure system safety.

The Spreading System has two versions, 1.0 and 2.0. They each have a different structure for the spinner disk. Compared with the Spreading System 1.0, the Spreading System 2.0 has a higher material delivery rate and can spread materials 360° around the aircraft.

1.1 In the Box

Spreading System×1	Spare Spinner Disk×2	M4x12 Screws×2	Fender×1 (Users have the option to purchase a fender with a screw pack to use with the Spreading System 2.0.)
			

1.2 Specifications

Items	Spreading System 1.0	Spreading System 2.0
Compatible Aircraft*	Agras T16, T20	
Spreader Weight	1.8 kg	2.1 kg (excluding fender)
Max Hopper Outlet Area	8.6 cm ²	32.3 cm ²
Compatible Material Diameter	0.5 - 5 mm	
Spread Tank Volume	20 L	
Spread Tank Internal Load	Using with the T16: 16 kg Using with the T20: 20 kg (note: the internal load of Japan version of T20 is 16 kg)	
Spreading Range	Varies according to material diameter, spinner disk rotating speed, hopper outlet size, and flying altitude. For best operating performance, it is recommended to adjust the corresponding variables to achieve a spreading range of 4 - 6 meters (for Spreading System 1.0) or 5 - 7 meters (for Spreading System 2.0).	

* The firmware of the aircraft must be the correct version for supporting the Spreading System. Check the release notes of the corresponding aircraft on the official DJI website.

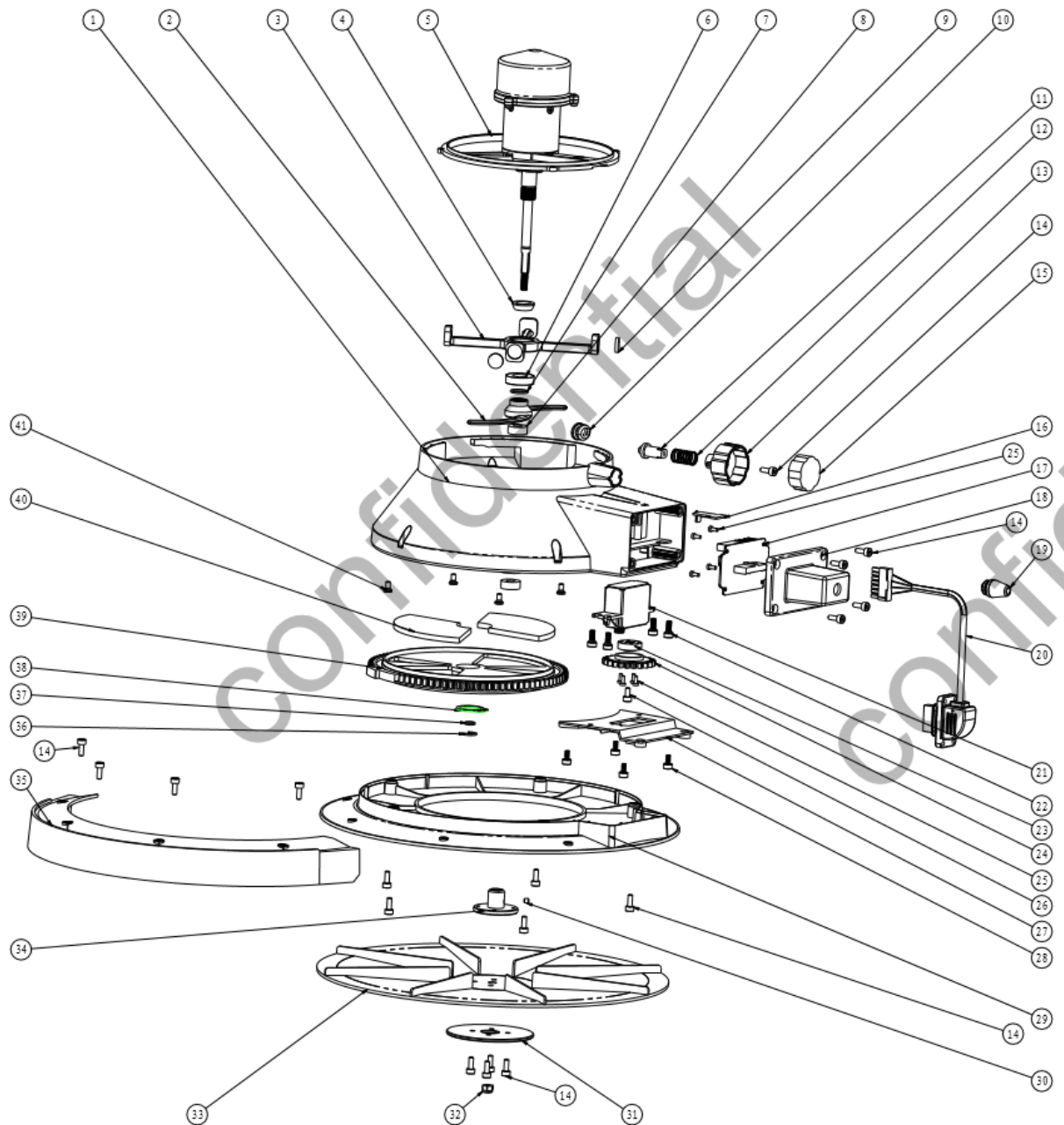
2 Product Comparison

	Spreading System 1.0	Spreading System 2.0	MG Spreading System 1.0
Compatible Aircraft	Agras T16, T20		MG-1S Series, MG-1S Advanced, MG-1P Series
Spreader Weight	1.8 kg	2.1 kg (Excluding Fender)	1.8 kg
Max. Hopper Outlet Area	8.6 cm ²	32.3 cm ²	8.6 cm ²
Compatible Material Diameter	0.5 - 5 mm	0.5 - 5 mm	0.5 - 5 mm
Spread Tank Volume	20 L	20 L	13 L
Spread Tank Internal Load	Using with the T16: 16 kg Using with the T20: 20 kg (note: the internal load of Japan version of T20 is 16 kg)		10 kg
Spreading Range*	4 - 6 m	5 - 7 m	4 - 6 m

*Varies according to material diameter, spinner disk rotating speed, hopper outlet size, and flying attitude. For best operating performance, it is recommended to adjust the corresponding variables to achieve a spreading range of 4-6 meters (for spreading system 1.0) or 5-7 meters (for spreading system 2.0).

3 BOM

3.1 Spreading System Main Module



Material List

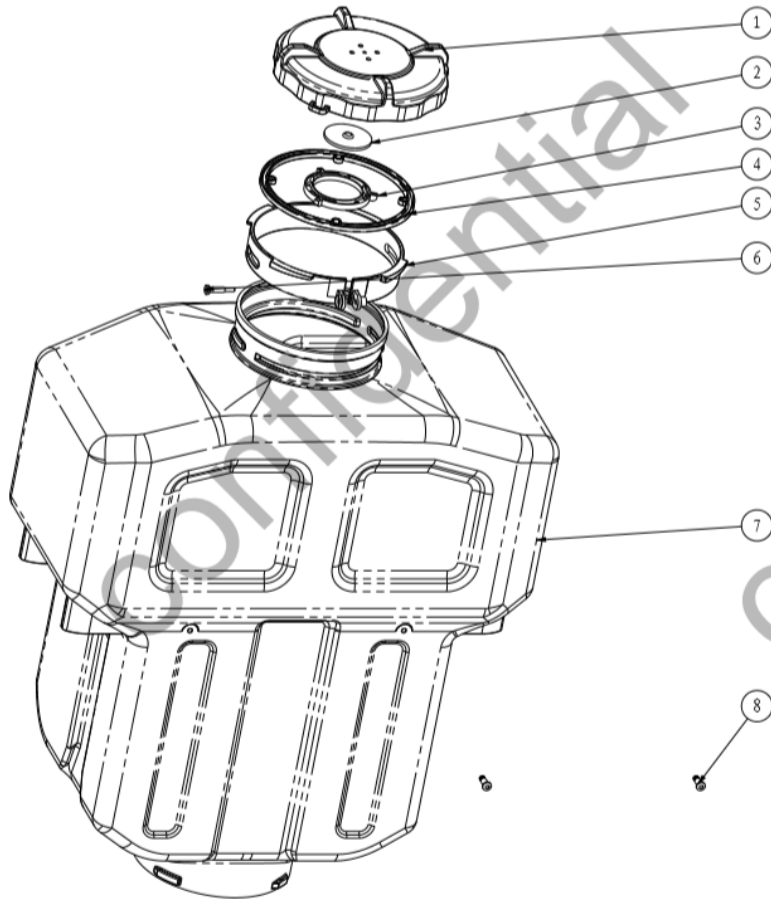
No.	Material Number	Material Name	Quantity
1&8	BC.AG.SS00028 0.01	Spreader Spreading Plate (With a Bearing) 2.0	1
2	YC.JG.QX00047 0.02	Muddler	1
4	YC.WJ.ZZ00097 1.02	Bearing Limit	1
5	BC.AG.SS00027 9.01	Reduction Gearbox 2.0	1

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3&6&9	BC.AG.SS00028 1.01	Spreading System Hall Sensor Module 2.0	1
7	YC.WJ.ZZ00097 0.01	Copper Sheet	1
10	YC.SJ.JG000180. 01	Motor Cable Silicone Rubber Ring	1
11	YC.WJ.ZZ00096 8.01	Spray Tank Lock	1
12	YC.WJ.TT00007 8.01	Fixing Spring	1
13	YC.SJ.WS00106 4.01	Spreader Lock Knob	1
14	YC.WJ.L00868.0 2	M30-HC080080-55-85	14
15	YC.BZ.XX000417 .01	Spreader Lock Knob EVA Foam	1
16	BC.TP.SS000067 .01	Spreader Hall Sensor Board	1
17	BC.AG.SS00030 5.01	Spreader Main Board2.0	1
18	YC.SJ.WS00106 3.02	PCB Box	1
19	YC.JG.MY00007 0.01	Cable Protective Silicone Rubber	1
20	YC.XC.XX000525 .03	Cable Connecting the Spreader and Spraying System Module	1
21&23& 24 &25&26	BC.AG.SS00034 6.01	Servo 2.0	1
22	YC.WJ.LL000074	T30-HC080065-60-15	4
27	YC.JG.ZS000275 .03	Dustproof Cover Plate	1
28	YC.WJ.LL000424 .01	T30-HC00600060-057030-0323-N	4
29	YC.JG.ZS000277 .03	Spinner Disk Upper Plate	1
30	YC.WJ.L00005.0 2	顶丝 Jack Screw 3*3	1
31	YC.WJ.ZZ00097 3.01	Spinner Disk Washer	1
32	YC.WJ.L00961.0 1	M4 Locknut	1
33	YC.JG.ZS000276 .02	Spinner Disk	1
34	YC.WJ.ZZ00096 9.01	Spreading Flange	1
36	YC.WJ.LP00001 5.01	Small Circlip	1
37	YC.WJ.ZZ00165 3.01	Small Bearing Gasket	1
39	YC.JG.ZS000278 .03	Large Gear	1

40	YC.JG.MQ00018 3.03	Teflon Pad	2
41	YC.WJ.LL000096	M30-HB050045-55-05	4

3.2 Spread Tank



Material List

No.	Material Number	Material Name	Quantity
1	YC.SJ.WS002418.04	Spread Tank Cover	1
2	YC.XJ.QT000200.02	Spread Tank Cover Air Outlet Sealing Pad	1
3	YC.SJ.WS002421.02	Spread Tank Cover Air Outlet cover	1
4	YC.XJ.QT000197.02	Spread Tank Cover Sealing Pad	1
5	YC.SJ.WS002423.02	Spread Tank Cover Ring	1
6	YC.WJ.LL000211.02	M30-HC02050080-070020-5103-Y	1
7	YC.SJ.WS003565.03	Spread Tank	1

8	YC.ST.LL900005.02	M40-HP01400040-068020-0003-Y	4
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4 Disassembly and Assembly & Notices

4.1 Tool

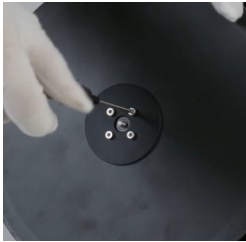
Tool	Specifications/Picture
3.0mm Hexagonal Socket Screwdriver	 A 3.0mm hexagonal socket screwdriver with a total length of 95.5mm and a 6mm hexagonal base.
2.5mm Hexagonal Socket Screwdriver	 A 2.5mm hexagonal socket screwdriver with a total length of 91mm.
2.0mm Hexagonal Socket Screwdriver	 A 2.0mm hexagonal socket screwdriver with a total length of 81.5mm.
1.5mm Hexagonal Socket Screwdriver	 A 1.5mm hexagonal socket screwdriver with a total length of 78mm.
Metal Tweezers	 A pair of metal tweezers with a total length of 120mm and a 10mm tip width.
Anti-Static Tweezers	 A pair of anti-static tweezers with a black handle.
7 mm Sleeve	 A 7mm sleeve with a green handle and a metal shaft.
Soldering Iron	
T4 Torx Screwdriver	

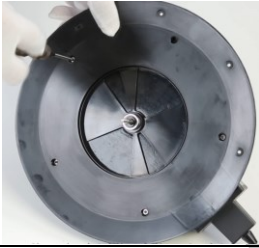
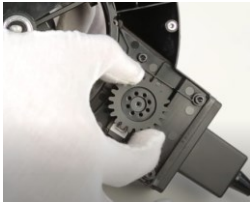
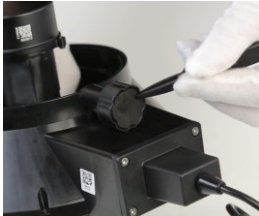
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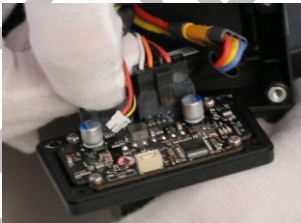
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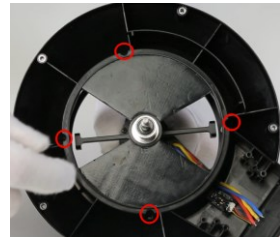
4.2 Disassembly Procedures

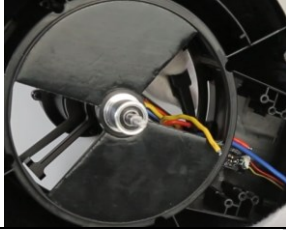
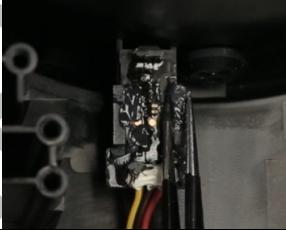
4.2.1 Disassembling the Spreading System Main Module

No.	Steps	Picture	Assembly Notices
1	Pull out the spreader lock knob, and rotate it 90 degrees.		
2	Rotate the spreader to separate the spread tank and spreader.		
3	Remove the M4 nut on the spinner disk with the sleeve.		
4	Unscrew the four screws securing the spinner disk's washer, then remove the washer.		
5	Remove the spinner disk.		

6	Loosen the set screw securing the flange, then remove the flange.		
7	Remove the five screws securing the spinner disk's upper plate, then remove the upper plate.		
8	Rotate the small gear counterclockwise to its limit and remove the screw securing the servo disk.		
9	Remove the four screws securing the small gear, then remove the large gear.		
10	Remove the small gear and servo disk.		
11	Remove the EVA foam on the spreader lock knob.		

12	Remove the screw securing the spreader lock knob, then remove the spreader lock knob.		
13	Remove the lock and spring as shown.		
14	Remove the four screws securing the PCB box.		
15	Disconnect all the cables on the main board.		
16	Remove the four screws securing the main board.		
17	Push out the cable connecting the spreader and spraying module. Then separate the main board and cable.		Remember to push out the cable from the main board. Do not pull the cable directly.
18	Remove the four screws securing the dustproof cover plate, then remove the cover plate.		

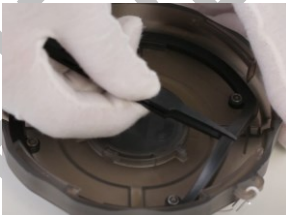
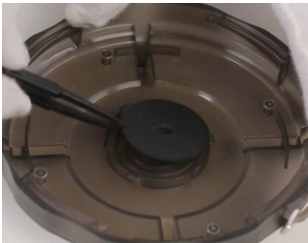
19	Remove the four screws securing the servo, then remove the servo.		Ensure that the servo cable is on the other side of the Hall sensor board when mounting the servo.
20	Remove the ESC board.		
21	Separate the sleeve and motor cables' connector. Desolder the connector with the electronic soldering iron.		Pay attention to the cable sequence when soldering.
22	Remove the sleeve.		
23	Remove the four screws securing the spinner disk and reduction gearbox.		
24	Remove the small circlip with the circlip pliers.		

25	Remove the small bearing's gasket.		
26	Pull out the three motor cables.		
27	Rotate the reduction gearbox to an angle to separate the gearbox and spinner disk.		
28	Clean off the black glue on the Hall sensor board with the tweezers and remove the board.		Do not damage the component when cleaning off the black glue with the tweezers.
29	Rotate the muddler clockwise to remove it.		
30	Remove the copper sheet, Hall sensing module, and bearing limit.		

31	Remove the reduction gearbox's protective silicone.		
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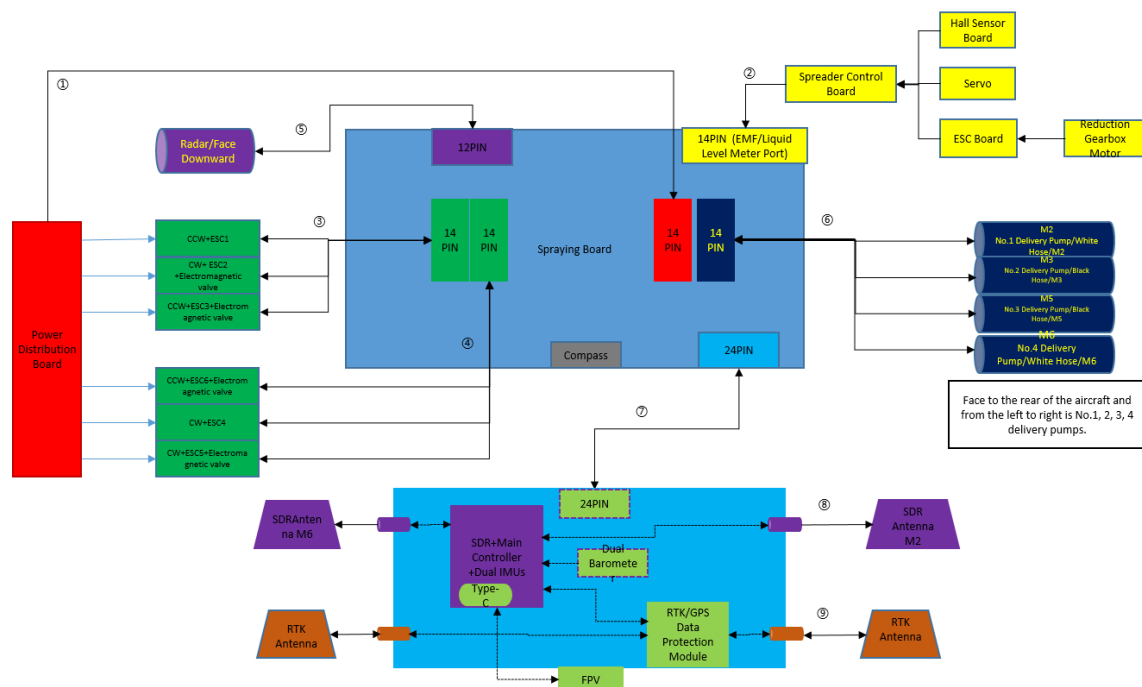
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4.2.2 Disassembling the Spread Tank

No.	Steps	Picture	Assembly Notices
1	Remove the spread tank's cover.		
2	Remove the long screw securing the spread tank garter spring. Separate the cover and garter spring as shown.		
3	Remove the cover's sealing ring.		
4	Rotate the rotatable air outlet cover to remove it.		
5	Remove the sealing pad.		

5 Principle

5.1 Connection Diagram and Boards Function



Using the connection between T20 and spreader as an example, the spreader is connected to the aircraft via the EMF/Liquid Level port on the spraying system.

The spreader includes five parts: the spray control board, Hall sensor board, servo, ESC board, and reduction gearbox motor.

1. The spray control board is the core module of the spreader and used to integrate the Hall sensor signal, ESC signal, and servo signal.
2. The Hall sensor board is a detection sensor for the gearbox motor rotation angle. It detects the angular position of the gearbox motor axis.
3. The servo is a control system which, as the hopper outlet angle change continually, controls the gearbox motor rotation angle.
4. The ESC board is a drive module for driving the gearbox motor rotation angle.
5. The reduction gearbox motor controls the hopper gate, driving the hopper outlet rotation and controlling the spreading amount.

6 Damage Assessment and Repair Guide

Module	Malfunction	Troubleshooting	Solution
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Control Board	Unable to be updated	1. Check whether DJI Assistant 2 for MG is the latest version.	Download the latest version of DJI Assistant 2 for MG from the official website.
		2. Replace the aircraft and update again. If the update is successful, the aircraft is malfunctioning. Then replace the spraying module and aerial-electronics system module to determine the corresponding defective component.	Replace the aerial-electronics system module or spraying module.
		3. Check whether the spreader control cable is damaged.	Replace the cable.
		4. Use the functioning control board to check whether the control board is damaged.	Replace the spreader control board.
Hall Sensor Board	A warning stating that the tank is empty is shown while the tank is not empty/there is no warning when the T16 liquid tank is empty.	1. The system needs to be calibrated again.	Perform calibration in the app again.
		2. Check whether the Hall sensor cable is loose or broken.	Ensure that the cable is functioning.
		3. Replace the Hall sensor board to check whether it is damaged.	Replace the Hall sensor board.
		4. Replace the control board to check whether it is damaged.	Replace the control board.
		5. Check whether the bearing is obstructed, which would cause the motor to drive the Hall sensor module to rotate.	Separate the bearing and clean it. If the issue cannot be resolved, replace the Hall sensor module.
Servo	The hopper outlet does not rotate.	1. Check whether the large and small gears are obstructed by debris.	Clean off the debris.
		2. Check whether the servo cable on the control board is loose.	Ensure that the connection is normal.
		3. Replace the servo to check whether it is damaged.	Replace the servo.
		4. Check whether the spreading system control cable on the control board is loose.	Replace the spreading system control cable.
		5. Replace the control board to check whether it is damaged.	Replace the control board.
	The hopper outlet cannot be completely opened or closed.	1. The system needs to be calibrated again.	Perform calibration in the app again.
		2. Check whether the large gear is obstructed by debris.	Clean off the debris.
		3. Check whether the large gear is damaged.	Replace the large gear.
		4. Check whether the small gear screws are loose.	Tighten the screws.
ESC Board/Reduction Gearbox Module	The spinner disk does not rotate.	1. Check whether the spinner disk is obstructed by debris.	Clean off the debris.
		2. Check whether the aircraft and spreading system firmware is the latest version.	Update the firmware.

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		3. Check whether any of the spreading flange's jack screws are loose.	Tighten the jack screws.
		4. The motor shaft on the gearbox is rusted.	Replace the gearbox module.
		5. Check whether the soldering part of the ESC cable on the gearbox is damaged or comes off.	Resolder the ESC cable.
		6. Replace the ESC board to check whether it is damaged.	Replace the ESC board.
		7. Check whether the spreading system control cable is damaged.	Replace the spreading system control cable.
		8. Replace the control board to check whether it is damaged.	Replace the control board.
		9. Replace the gearbox module to check whether it is damaged.	Replace the gearbox module.
	The speed of the spinner disk is slow.	1. Check whether the spinner disk is obstructed by debris.	Clean off the debris.
		2. Check whether the bearing on the spinner disk is obstructed by debris.	Replace the spinner disk.
		3. Replace the ESC board to check whether it is damaged.	Replace the ESC board.
		4. Replace the control board to check whether it is damaged.	Replace the control board.
		5. Replace the gearbox module to check whether it is damaged.	Replace the gearbox module.

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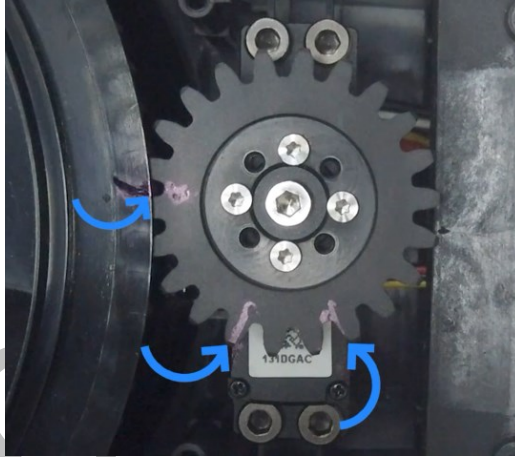
7 Test Instructions

Module	No.	Test Item	Content
Spreader	1	Product Calibration Test	Precision Calibration

Note: Ensure that the servo and small gear have been calibrated by the manufacturer. Please mount them following the sequence below to avoid affecting the level of precision. (Do not rotate the small gear during installation).

Scenario 1:

1. When mounting the servo (including the small gear), after the servo is connected to the control board, connect the spreader to the aircraft and power on the aircraft. Then the servo will be automatically turned on. The servo on the spreader will be restarted and the servo rotation will be limited. Then secure the servo on the spreader with the screws.
2. Mount the large gear (ensure that the hopper outlet is closed). Then mark a line on the large and small gears. Then expand the two marked lines on the small gear until reaching the external shell. In the picture below, the blue arrows point to how to mark the lines. (Lines on the large gear and small gear: Used to mark the relative position of the small gear and large gear; two lines on the small gear and external shell: Used to mark the small gear's position.)



3. Remove the large gear, loosen the screws on the small gear, and then remove the small gear. Mount the dustproof cover and tighten it. Then mark the lines on the small gear and large gear based on the previous step 2.
4. Power off the aircraft and finish the spreading system assembly. Then perform calibration in the app.

Scenario 2:

Before mounting the servo (including the small gear), the servo and small gear are separated, causing the original relative position of the servo and small gear to have changed.

1. Mount the servo. After the servo is connected to the control board, connect the spreader to the aircraft.
2. Mount the large gear (ensure that the hopper outlet is closed), and then mount the small gear to the servo. Ensure that the small gear and large gear are matched. Then check that the screw hole on the small gear is aligned with the screw hole on the servo. If yes, then tighten the small gear in place with the screws. If the screw holes are not aligned, adjust the small gear (ensure that the large gear is not moved) until the screw holes are aligned. Then tighten the screws on the small gear. Afterwards, finish the spreading system assembly, and then perform calibration in the app.



7.1 Test Items

			Is Test Required After Component Replacement							
Module	No.	Test Item	Control Board	ESC Board	Servo	Gear box	Hall Sensor Board	Spinner Disk	Large Gear	Small Gear
Spreader	1	Product Calibration Test	√	√	√	√	√	√	√	√